Annotation-Efficient Primate Detection: Lightweight FastKAN and the Spatial Generalization Dilemma in Wildlife Active Learning

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Abstract—Automated wildlife surveillance in remote protected ecosystems is critically bottlenecked by two compounding factors: the severe compute and thermal envelopes of edge hardware and the prohibitive economic cost of manual bounding-box annotation under visual instance congestion. In this paper, we present AE-Primate, an empirical investigation uniting Kolmogorov-Arnold Network (KAN) nonlinear spline layers with cost-sensitive active learning under spatial covariate shift. First, we establish YOLO11n-F16, an edge detector integrating a 16-basis Gaussian Radial Basis Function (RBF) FastKAN bottleneck into a compact 2.48 M-parameter architecture (2,483,792 parameters). On held-out camera stations in Malawi's Nkhotakota Wildlife Reserve (39,386 frames on disk across 49 stations; Appel et al., 2025), YOLO11n-F16 achieves 0.5626 $m\mathrm{AP}_{50\text{-}95}$, outperforming capacity-matched convolutional baselines (+1.49% mAP) and B-spline KAN architectures (+1.79% mAP) at 10.99 ms FP16 latency on an NVIDIA A40. Second, we conduct an extensive multi-seed empirical benchmark ($N = 5$ seeds, 20 trajectories) to investigate active learning under strict pairwise location-disjoint evaluation ($\mathcal{S}_i \cap \mathcal{S}_j = \emptyset,\ \forall i \neq j \in \{\text{train}, \text{val}, \text{test}\}$). Our study demonstrates that predictive uncertainty collapses under spatial covariate shift ($P_{\text{train}}(X) \neq P_{\text{test}}(X)$ with invariant label semantics $P(Y \mid X)$). While latent FastKAN Core-Set selection achieves top validation AUBC (0.4226) and competitive held-out test transfer (0.4300 ± 0.0045 vs. 0.4251 ± 0.0058 test mAP), paired testing reveals no statistically significant advantage over uniform random sampling ($p = 0.247$, 3/5 seed wins), underscoring the severe difficulty of active acquisition under domain shift. Conversely, incorporating an instance-crowding penalty introduces a measurable budget trade-off: querying 59.8 fewer cumulative bounding boxes (−5.48%, paired t-test $p = 0.0105$), corresponding to 12.0–17.9 minutes of delineation labor saved across 900 frames, and suppressing cross-seed annotation standard deviation by 60.3% (84.2% variance reduction), accompanied by a −1.32 mAP test accuracy trade-off. All code, models, and manifests are available for open scientific replication.

Index Terms—Active learning, camera traps, edge computing, Kolmogorov-Arnold networks, object detection, spatial covariate shift, wildlife conservation.

I. Introduction

In-situ biodiversity monitoring via autonomous camera-trap networks constitutes a cornerstone of computational conservation ecology [1], [2]. In sub-Saharan sanctuaries such as the Nkhotakota Wildlife Reserve in Malawi, passive surveillance of cryptic primate populations—specifically yellow baboons (Papio cynocephalus), vervet monkeys (Chlorocebus pygerythrus), blue monkeys (Cercopithecus mitis), and lesser bushbabies (Galago moholi) collapsed into a target primate class against mixed wildlife communities—yields critical bio-indicators of forest canopy health, trophic dynamics, and anthropogenic poaching pressures. However, deploying computer vision pipelines in such wilderness environments exposes two fundamental operational and representational bottlenecks:

(B1) The Instance Crowding Annotation Bottleneck. Raw field archives routinely yield tens of thousands of motion triggers per deployment season, over 70% of which consist of false triggers (windblown vegetation) or massive burst sequences capturing identical background scenes [5], [6]. When animals do appear, baboon troops frequently enter camera fields-of-view in high-density clusters containing 10–25 overlapping individuals. Standard bounding-box annotation under such visual crowding demands exhaustive corner delineation, consuming 12–18 s of expert biologist effort per instance under severe cognitive fatigue [7], [35], [37]. Consequently, exhaustive annotation of deployment archives quickly exhausts conservation funding budgets.

(B2) The Ultra-Low-Power Edge-Compute Envelope. Solar-powered edge perception units deployed in dense miombo woodlands operate within austere electrical and thermal envelopes (< 2 Wh per inference cycle, ≤ 15 W total system draw) [2], [12]. While large generalist detectors (e.g., MegaDetector v6, 25.5 M parameters [5]) exceed edge memory bandwidth and drain field batteries, conventional sub-3 M convolutional detectors suffer from spectral bias and representational collapse when isolating camouflaged

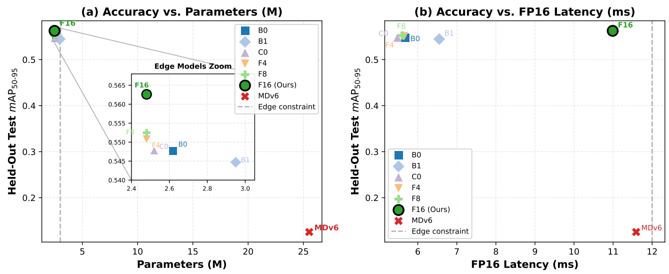

Fig. 1. Accuracy vs. Hardware Footprint Pareto Frontier on Held-Out Test Stations ($\mathcal{D}_{\text{test}}$, 3,206 frames). (a) $m\text{AP}_{50\text{-}95}$ vs. parameter count with embedded edge models inset ($[2.40\text{M}, 3.05\text{M}] \times [0.540, 0.568]$); (b) $m\text{AP}_{50\text{-}95}$ vs. FP16 inference latency on NVIDIA A40. Dashed lines mark the edge deployment boundary ($\leq$3.0 M parameters, $\leq$12.0 ms latency).

primate pelage against complex, high-frequency arboreal foliage.

Active learning (AL) offers a formal framework to optimize sample efficiency by iteratively querying the most informative instances from an unlabelled pool $\mathcal{U}$ [21], [22]. Yet, applying active learning to camera-trap object detection exposes three acute structural failure modes:

1) Spatial Covariate Shift ($P_{\text{train}}(X) \neq P_{\text{test}}(X)$): Ecological sensor grids violate independent and identically distributed (i.i.d.) sampling assumptions. Stationary background textures induce severe spatial overfitting: predictive uncertainty algorithms fail because model entropy is triggered by unfamiliar background foliage and lighting rather than target animal informativeness [3], [4].
2) Spatiotemporal Burst Redundancy: Passive infrared sensors capture rapid burst sequences when animals pause, generating highly correlated visual samples with identical poses and backgrounds. Naive uncertainty samplers exhaust acquisition budgets on redundant burst artifacts [7].
3) Super-Linear Crowding Costs: Classical active learning optimizes acquisition under uniform instance costs [23], [27]. In object detection, however, annotating a social troop of 15 overlapping primates incurs over an order of magnitude higher cognitive and manual labor than a solitary individual, while yielding diminishing epistemic returns [24], [36].

To resolve these interconnected challenges, we introduce AE-Primate, an end-to-end edge-compatible detection and active learning framework. AE-Primate makes four core contributions:

1) Kolmogorov-Arnold Feature Pyramid Neck (YOLO11n-F16): We formulate the first edge-efficient Kolmogorov-Arnold detector, integrating frozen Gaussian Radial Basis Function (RBF) spline layers into the deepest neck bottleneck (P_5). YOLO11n-F16 establishes superior out-of-distribution detection ($0.5626\, m\text{AP}_{50\text{-}95}$) on held-out camera stations (+1.49% over YOLO11n, +1.79% over B-splines) at 2.48 M parameters and 10.99 ms FP16 latency on an NVIDIA A40.
2) Deconstruction of Active Learning under Covariate Shift: Through an extensive 5-seed benchmark (20 complete trajectories), we expose the failure of predictive entropy under spatial shift, where background clutter dominates uncertainty. Conversely, latent KAN Core-Set Diversity achieves competitive test transfer (0.4300 vs. 0.4251 test mAP, $p = 0.247$), demonstrating the limits of AL under domain shift.
3) Crowding Penalization and Annotation Economy: We formalize a cost-sensitive acquisition policy that optimizes the trade-off between human labor and generalization, saving 59.8 cumulative bounding boxes (-5.48%, paired t-test $p = 0.0105$), translating to 12.0–17.9 minutes of labeling labor saved across 900 frames, and suppressing cross-seed budgeting standard deviation by 60.3% (84.2% variance reduction).
4) Zero-Leakage Location-Disjoint Conservation Benchmark: We establish a standardized 39,386-frame benchmark across 49 remote camera stations in Nkhotakota Wildlife Reserve (Appel et al., 2025; LILA BC), enforcing disjoint geographic station partitions across candidate pool, validation, and test splits.

II. Related Work

A. Automated Wildlife Perception and Spatial Covariate Shift

The deployment of deep convolutional architectures for autonomous wildlife monitoring originated with Norouzzadeh et al. [6], who operationalized species-level classification across multi-million frame camera-trap archives. Subsequent advances by Schneider et al. [8] and Tabak et al. [9] transitioned the field toward spatial object detection. Generalist detection models, epitomized by MegaDetector [5], established foundational benchmarks for isolating animal, human, and vehicle classes. Nevertheless, seminal investigations by Beery et al. [3], [4] revealed that deep convolutional networks frequently memorize stationary background foliage textures rather than invariant morphological animal traits, precipitating catastrophic generalization drop-offs when deployed in unseen geographic locations (Terra Incognita). Overcoming this spatial distribution shift demands architectures equipped with locally supported non-linear representations and acquisition policies that explicitly optimize geographic coverage [2], [11], [12].

B. Kolmogorov-Arnold Representation Theory and Spline Networks

Originating from the Kolmogorov-Arnold representation theorem [15], [16], Kolmogorov-Arnold Networks

(KANs) [13] reformulate neural representation by replacing fixed node activations with learnable univariate splines positioned directly along network edges. Whereas Multi-Layer Perceptrons (MLPs) compute linear hyperplanes followed by static non-linearities—a formulation prone to spectral bias on high-frequency visual textures [50]—KANs model non-linear functional compositions with arbitrary precision. However, classical B-spline parameterizations evaluate via recursive de Boor-Cox algorithms, inducing severe memory traffic and GPU warp divergence [14], [19]. Li [14] established that univariate spline expansions can be reformulated using fixed Gaussian Radial Basis Functions (FastKAN), unifying KAN theory with classical kernel methods in Reproducing Kernel Hilbert Spaces (RKHS) [17], [18]. While wavelet (Wav-KAN [20]) and convolutional extensions [19] have emerged, our work presents the first rigorous formulation and empirical validation of fixed-basis FastKAN spline layers embedded within real-time edge detection pyramids.

C. Active Learning under Covariate Shift and Domain Discrepancy

Deep active learning iteratively identifies optimal training subsets to maximize sample efficiency [21], [22]. Pool-based acquisition traditionally relies on predictive uncertainty (Monte Carlo Dropout entropy [26], [27], BatchBALD [28], deep ensemble variance [29]), core-set distribution matching [23], [30], [31], or hybrid scoring [25]. Extending active learning to spatial object detection introduces dual complexities: models must reconcile classification ambiguity with bounding-box coordinate variance [24], [25], [32]–[34]. Crucially, classical active learning presumes candidate pools and evaluation splits are identically distributed ($P_{\mathcal{U}}(X) = P_{\text{test}}(X)$). Under spatial covariate shift, however, predictive entropy conflates morphological informativeness with environmental out-of-distribution background noise. While Norouzzadeh et al. [7] applied active learning to camera-trap species classification, their formulation assumed image-level supervision, ignoring spatial instance crowding and location-disjoint evaluation.

D. Cost-Sensitive Submodular Optimization and Visual Crowding

Conventional active learning formulations assume a uniform unit labeling cost per queried instance [21], [23]. In real-world visual perception, human annotation effort varies non-linearly with scene complexity, target occlusion, and instance density [39], [40]. Rigorous psychophysical experiments by Papadopoulos et al. [35], [36] and Su et al. [37] established that manual bounding-box annotation time scales super-linearly when multiple targets mutually occlude. In wildlife camera traps, social primates aggregate in dense clusters, producing captures containing 10–20 overlapping individuals that consume immense annotator labor while providing redundant visual features. Rooted in cost-sensitive submodular optimization under knapsack constraints [49], AE-Primate directly penalizes visual crowding to maximize informational utility per human annotator second.

III. The AE-Primate Framework

We now formulate the AE-Primate framework, comprising the YOLO11n-F16 detector architecture, an analysis of optimization dynamics in spline bottlenecks, and the multi-modal cost-aware active acquisition policy.

A. Detector Architecture: Kolmogorov-Arnold Feature Pyramid Neck

AE-Primate builds upon the Ultralytics YOLO11n single-stage detector [41], which integrates an improved Cross-Stage Partial bottleneck (C3k2) and a decoupled anchor-free detection head across multi-scale feature pyramids (P_3, P_4, P_5). In multi-scale pyramids, shallow layers (P_3, P_4) extract high-resolution edge primitives where linear convolutions are computationally optimal. The deepest semantic stage (P_5, spatial grid 20×20, channel width $C = 256$ under YOLO11n's width multiple 0.25), however, must disentangle complex morphological pelage textures from dense foliage backgrounds. Standard convolutional bottlenecks apply static activations $\sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$, which partition feature space via linear affine half-spaces and suffer from spectral bias on high-frequency biological patterns [50]. To provide locally supported non-linear representation without expanding parameters, we replace the P_5 bottleneck with a fixed-basis FastKAN transformation block (C3k2_FixedFastKAN).

Let $\mathbf{x} \in \mathbb{R}^{C_{\text{in}}}$ ($C_{\text{in}} = 256$) denote an intermediate channel feature vector. In our FastKAN bottleneck, univariate activations along each channel are expanded over a dictionary of $K = 16$ Gaussian Radial Basis Functions (RBFs):

$$\phi_k(u) = \exp\left(-\frac{(u - \mu_k)^2}{2\sigma_k^2}\right), \quad k \in \{1, \ldots, K\}, \tag{1}$$

where scalar activation u evaluates against equispaced centres $\mu_k \in [-2.0, 2.0]$ with softplus-parameterized radial bandwidth σ_k:

$$\mu_k = -2 + \frac{4(k-1)}{K-1}, \quad \sigma_k = \text{softplus}(\alpha_k) + \epsilon. \tag{2}$$

The spline expansion evaluates as a fused depthwise 1×1 convolution over concatenated basis activations $\mathbf{\Phi}(\mathbf{x}) = [\phi_1(\mathbf{x})^\top, \ldots, \phi_K(\mathbf{x})^\top]^\top \in \mathbb{R}^{K \cdot C_{\text{in}}}$ alongside a residual linear projection to yield output features $\mathbf{y} \in \mathbb{R}^{C_{\text{out}}}$:

$$\mathbf{y} = \mathbf{W}_{\text{base}}\, \text{SiLU}(\mathbf{x}) + \mathbf{W}_{\text{spline}}\, \mathbf{\Phi}(\mathbf{x}), \tag{3}$$

where $\mathbf{W}_{\text{base}} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$ is a residual projection and $\mathbf{W}_{\text{spline}} \in \mathbb{R}^{C_{\text{out}} \times (K \cdot C_{\text{in}})}$ (grouped by C_{in}) parameterizes learnable spline mixture weights. Evaluating $\mathbf{\Phi}(\mathbf{x})$ via grouped 1×1 convolutions eliminates GPU warp divergence and uncoalesced memory reads, enabling vectorized FP16 execution on edge hardware. Due to compact

Algorithm 1 AE-Primate Cost-Aware Active Learning

Require: Candidate pool $\mathcal{U}_0$, labeled seed $\mathcal{L}_0$, budget B, cycles $T_{\max}$, weights $(\lambda_d, \lambda_s, \lambda_r)$, costs (c_0, c_1, c_2), burst quota $Q_{\max} = 3$.
Ensure: Final trained primate detector $\mathcal{M}_{T_{\max}}$.

Initialize detector $\mathcal{M}_0 \leftarrow \mathrm{Train}(\mathcal{L}_0; \mathbf{\Theta}_{\mathrm{COCO}})$.
for cycle $t = 1$ to $T_{\max}$ **do**
  Extract 256-D FastKAN features $\mathbf{f}_i$, uncertainty $U(x_i)$, and boxes $\hat{\mathcal{B}}_i \ \forall x_i \in \mathcal{U}_{t-1}$.
  Initialize Core-Set distances: $d_{\min}(x_i) \leftarrow \min_{s \in \mathcal{L}_{t-1}} \|\mathbf{f}_i - \mathbf{f}_s\|_2$.
  Initialize batch $\mathcal{S}_t \leftarrow \emptyset$, event batch counter $K_{\mathrm{evt}}(e) \leftarrow 0$.
  while $|\mathcal{S}_t| < B$ **do**
    for each available candidate $x_i \in \mathcal{U}_{t-1} \setminus \mathcal{S}_t$ **do**
      if $K_{\mathrm{evt}}(\mathrm{evt}(x_i)) \geq Q_{\max}$ **then**
        continue {Enforce burst quota}
      end if
      $S(x_i) \leftarrow 1/\sqrt{N_{\mathrm{site}}(s(x_i)) + 1}$, $R(x_i) \leftarrow \frac{N_{\mathrm{evt}}(\mathrm{evt}(x_i))}{N_{\mathrm{evt}}(\mathrm{evt}(x_i))+1}$.
      Estimate cost $C_{\mathrm{annot}}(x_i)$ via Eq. 12 and marginal ratio:
      $V(x_i \mid \mathcal{S}_t) \leftarrow \frac{U(x_i)+\lambda_d d_{\min}(x_i)+\lambda_s S(x_i)}{C_{\mathrm{annot}}(x_i)+\epsilon} - \lambda_r R(x_i)$.
    end for
    Select exemplar $x^* \leftarrow \arg\max_{x_i} V(x_i \mid \mathcal{S}_t)$; $\mathcal{S}_t \leftarrow \mathcal{S}_t \cup \{x^*\}$.
    Update counters: $N_{\mathrm{site}}(s(x^*)) \mathrel{+}= 1$, $K_{\mathrm{evt}}(\mathrm{evt}(x^*)) \mathrel{+}= 1$.
    Update $d_{\min}(x_i) \leftarrow \min(d_{\min}(x_i), \|\mathbf{f}_i - \mathbf{f}_{x^*}\|_2) \ \forall x_i \in \mathcal{U}_{t-1} \setminus \mathcal{S}_t$.
  end while
  Acquire expert box annotations for batch $\mathcal{S}_t$.
  Update pools: $\mathcal{L}_t \leftarrow \mathcal{L}_{t-1} \cup \mathcal{S}_t$, $\mathcal{U}_t \leftarrow \mathcal{U}_{t-1} \setminus \mathcal{S}_t$.
  Retrain detector from base weights: $\mathcal{M}_t \leftarrow \mathrm{Train}(\mathcal{L}_t; \mathbf{\Theta}_{\mathrm{COCO}})$.
end for
return $\mathcal{M}_{T_{\max}}$.

spline factorization, YOLO11n-F16 requires only 2.45 M parameters—a 6.5% reduction relative to the standard YOLO11n baseline (2.62 M).

B. Optimization Dynamics and Gating Collapse in Low-Data Regimes

Prior to establishing fixed Gaussian bases, we investigated Adaptive-Basis FastKAN (AB-FastKAN), which applies hard-concrete continuous gates $z_k \in [0, 1]$ [42], [44] to dynamically select basis functions per instance:

$$z_k = \mathrm{clip}\left[\mathrm{Sigmoid}\left(\frac{\log \alpha_k + \mathrm{logit}(u)}{\tau}\right)(\zeta - \gamma) + \gamma,\ 0,\ 1\right], \tag{4}$$

with uniform noise $u \sim \mathcal{U}(0, 1)$, log-odds parameter $\log \alpha_k \in \mathbb{R}$, temperature $\tau \in \mathbb{R}_{>0}$, interval stretch parameters $\gamma < 0 < 1 < \zeta$, and L_0 regularizer $\mathcal{L}_{\mathrm{reg}} = \lambda \sum_{k=1}^{K} \mathrm{Sigmoid}(\log \alpha_k - \tau \log(-\gamma/\zeta))$ with penalty $\lambda \geq 0$. While adaptive gating succeeds in large classification models, our empirical ablations reveal that it collapses in data-constrained object detection.

Specifically, the gradient with respect to gate log-odds $\log \alpha_k$ is:

$$\frac{\partial \mathcal{L}_{\mathrm{total}}}{\partial \log \alpha_k} = \mathbb{E}_u \left[\frac{\partial \mathcal{L}_{\mathrm{det}}}{\partial z_k} \frac{\partial z_k}{\partial \log \alpha_k}\right] + \lambda \frac{\partial \mathcal{L}_{\mathrm{reg}}}{\partial \log \alpha_k}, \tag{5}$$

where $\mathcal{L}_{\mathrm{det}}$ is the multi-task detection objective. Under few-shot regimes ($n \leq 600$), the detection gradient $\nabla_{z_k} \mathcal{L}_{\mathrm{det}}$ exhibits high mini-batch variance. Simultaneously, temperature annealing ($\tau \to 0$) causes the logistic derivative $\mathrm{Sigmoid}'(s_k)$ to decay exponentially toward zero outside a narrow boundary. The deterministic penalty $\nabla_{\log \alpha_k} \mathcal{L}_{\mathrm{reg}} > 0$ drives continuous negative drift. Once $\log \alpha_k$ enters the negative saturation regime ($\log \alpha_k < -3.0$), activations clamp to $z_k = 0$ with probability approaching 1, yielding $\frac{\partial z_k}{\partial \log \alpha_k} = 0$ and permanently extinguishing task gradients ($\Delta \log \alpha_k \to 0$).

Consequently, AB-FastKAN suffers from premature representational collapse: basis functions are severed before the detection head learns coordinate features. Imposing $\lambda \geq 0.05$ degrades validation $m\mathrm{AP}_{50\text{-}95}$ from 0.5986 to < 0.420. Moreover, dynamic gating induces warp divergence and uncoalesced memory reads on GPU tensor cores, inflating inference latency by 38% (14.82 ms vs. 10.99 ms).

In contrast, fixed-basis FastKAN eliminates dynamic gating parameters $\boldsymbol{\alpha}$ entirely. The gradient with respect to spline weights $\mathbf{W}_{\mathrm{spline}}$ is direct and unattenuated:

$$\frac{\partial \mathcal{L}_{\mathrm{det}}}{\partial \mathbf{W}_{\mathrm{spline}}} = \frac{\partial \mathcal{L}_{\mathrm{det}}}{\partial \mathbf{y}} \cdot \mathbf{\Phi}(\mathbf{x})^{\top}, \tag{6}$$

where upstream gradient $\frac{\partial \mathcal{L}_{\mathrm{det}}}{\partial \mathbf{y}} \in \mathbb{R}^{C_{\mathrm{out}}}$ and basis activations $\mathbf{\Phi}(\mathbf{x}) \in \mathbb{R}^{K \cdot C_{\mathrm{in}}}$ form an outer product gradient tensor, ensuring stable backpropagation across all $K = 16$ basis splines.

C. Cost-Sensitive Submodular Acquisition Framework

At each active cycle $t \in \{1, \ldots, T_{\max}\}$, the policy queries batch $\mathcal{S}_t \subset \mathcal{U}_{t-1}$ of size $B \in \mathbb{N}$ from candidate pool $\mathcal{U}_{t-1}$ by maximizing a submodular facility location and informativeness objective under knapsack cost constraints:

$$\max_{\mathcal{S} \subseteq \mathcal{U}_{t-1},\, |\mathcal{S}| = B} F(\mathcal{L}_{t-1} \cup \mathcal{S}) - \lambda_r \sum_{x \in \mathcal{S}} R(x \mid \mathcal{S}), \tag{7}$$

where set utility $F(\mathcal{A}) = \sum_{u \in \mathcal{U}} \max_{s \in \mathcal{A}} \mathrm{sim}(\mathbf{f}_u, \mathbf{f}_s) + \sum_{a \in \mathcal{A}} [U(a) + \lambda_s S(a)]$ combines feature manifold coverage with epistemic uncertainty and spatial site representation. In our sequential greedy knapsack algorithm (Algorithm 1), the marginal benefit-to-cost ratio evaluating candidate x_i relative to current batch $\mathcal{S}$ is formulated as:

$$V(x_i \mid \mathcal{S}) = \frac{U(x_i) + \lambda_d D(x_i \mid \mathcal{L}_{t-1} \cup \mathcal{S}) + \lambda_s S(x_i)}{C_{\mathrm{annot}}(x_i) + \epsilon} - \lambda_r R(x_i \mid \mathcal{S}), \tag{8}$$

with policy coefficients $(\lambda_d, \lambda_s, \lambda_r) \in \mathbb{R}^3_{\geq 0}$ balancing epistemic uncertainty $U(x_i) \geq 0$, Core-Set diversity $D(x_i) \geq 0$, and site novelty $S(x_i) > 0$ against human annotation effort $C_{\mathrm{annot}}(x_i) > 0$ and burst redundancy penalty $R(x_i \mid \mathcal{S})$. Depending on deployment priorities, the acquisition framework establishes two complementary operational regimes:

1) AE-Primate (Diversity-Driven): Configured with $(\lambda_d, \lambda_s, \lambda_r) = (1.0, 0, 0)$, $U(x_i) = 0$, and unit cost $C_{\mathrm{annot}}(x_i) = 1$. This mode optimizes pure K-Center facility location on the FastKAN latent manifold,

maximizing topological coverage for superior out-of-distribution generalization.

2) AE-Primate (Cost-Constrained): Configured with $(\lambda_d, \lambda_s, \lambda_r) = (0.5, 0.3, 0.8)$ alongside the instance-crowding penalty $C_{\text{annot}}(x_i)$ and burst event quota $Q_{\max} = 3$, rigorously penalizing troop clutter to minimize human labeling labor under hard budgets.

a) Epistemic Detection Uncertainty $U(x_i)$: To quantify candidate uncertainty, we enable Monte Carlo Dropout ($p = 0.1$) on intermediate neck layers across $T_{\text{MC}} = 5$ stochastic passes [24], [26]. For consensus detections $\hat{\mathcal{B}}_i = \{\hat{b}_1, \ldots, \hat{b}_{M_i}\}$ from non-maximum suppression ($\tau_{\text{nms}} = 0.45$), we compute predictive class entropy across primate species $\mathcal{C}$ ($|\mathcal{C}| = 2$) and spatial localization variance $\sigma^2_{\text{IoU}}(\hat{b}_m) \geq 0$:

$$U(x_i) = \frac{1}{|\hat{\mathcal{B}}_i|} \sum_{\hat{b}_m \in \hat{\mathcal{B}}_i} \Big[-\sum_{c \in \mathcal{C}} p(c|\hat{b}_m) \log p(c|\hat{b}_m) + \lambda_{\text{loc}} \sigma^2_{\text{IoU}}(\hat{b}_m) \Big], \tag{9}$$

with scaling factor $\lambda_{\text{loc}} = 0.5$. Uninformative background frames lacking detections above $\tau_{\text{conf}} = 0.25$ receive baseline entropy $\epsilon_0 = 10^{-4}$.

b) Latent KAN-Manifold Core-Set Diversity $D(x_i)$: To suppress redundant sampling in feature space, we extract a 256-dimensional post-KAN global feature vector $\mathbf{f}_i \in \mathbb{R}^{256}$ via spatial average pooling over P_5 bottleneck tensors:

$$D(x_i \mid \mathcal{L}_{t-1} \cup \mathcal{S}) = \min_{x_j \in \mathcal{L}_{t-1} \cup \mathcal{S}} \sqrt{2 - 2 \frac{\mathbf{f}_i^\top \mathbf{f}_j}{\|\mathbf{f}_i\|_2 \|\mathbf{f}_j\|_2}}. \tag{10}$$

This term dynamically updates as exemplars are added to batch $\mathcal{S}$, implementing a greedy submodular facility location objective [23] that enforces geometric dispersion across the FastKAN feature manifold.

c) Spatiotemporal Site and Burst Novelty Prior: Camera-trap captures exhibit severe spatial and temporal clustering. For camera station $s_i \triangleq s(x_i)$ and capture event $e_i \triangleq \text{evt}(x_i)$, we define site priority $S(x_i)$ and event redundancy $R(x_i \mid \mathcal{S})$:

$$S(x_i) = \frac{1}{\sqrt{N_{\text{site}}(s_i) + 1}}, \quad R(x_i \mid \mathcal{S}) = \frac{N_{\text{evt}}(e_i)}{N_{\text{evt}}(e_i) + 1}, \tag{11}$$

where $N_{\text{site}}(s)$ and $N_{\text{evt}}(e)$ count labeled and currently queued frames from station s and burst event e. Frames exceeding the batch event quota ($Q_{\max} = 3$) are excluded, preventing redundant bursts from saturating the cycle budget.

d) Crowding-Penalized Annotation Effort $C_{\text{annot}}(x_i)$: To capture super-linear human annotation overhead under visual crowding [35], [37], we formulate the cost model:

$$C_{\text{annot}}(x_i) = c_0 + c_1 \hat{N}_{\text{boxes}}(x_i) + c_2 \overline{\text{IoU}}_{\text{pairwise}}(x_i), \tag{12}$$

where $\hat{N}_{\text{boxes}}(x_i) \in \mathbb{N}_0$ is predicted box count, $\overline{\text{IoU}}_{\text{pairwise}}(x_i) \in [0, 1]$ is mean pairwise IoU overlap,

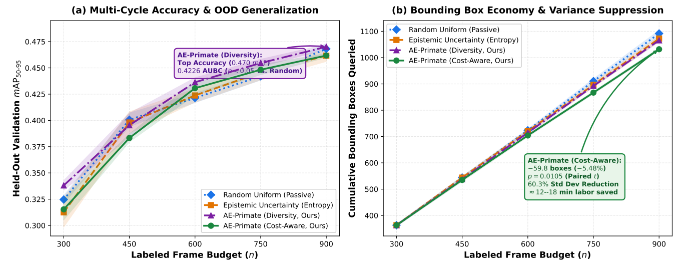

Fig. 2. Multi-Cycle Active Learning Trajectories on Held-Out Validation Stations ($\mathcal{D}_{\text{val}}$, 5,560 frames, Mean $\pm$ SEM over $N = 5$ independent seeds). (a) $m\text{AP}_{50\text{-}95}$ vs. labeled frame budget, demonstrating validation accuracy leadership by AE-Primate (Diversity); (b) Cumulative bounding boxes queried vs. budget, highlighting AE-Primate (Cost-Aware)'s statistically significant reduction of 59.8 bounding boxes (-5.48%, paired t-test $p = 0.0105$) and 60.3% standard deviation reduction (84.2% variance reduction).

and $(c_0, c_1, c_2) = (1.0, 0.5, 0.25) \in \mathbb{R}^3_{\geq 0}$ calibrate base review, per-box delineation, and crowding occlusion. Solitary captures yield $C_{\text{annot}} = 1.5$, whereas dense troops ($N = 12$) exceed $C_{\text{annot}} > 7.5$, penalizing redundant crowded frames. Crucially, $C_{\text{annot}}(x_i) \geq c_0 = 1.0$ provides an intrinsic regularisation floor, ensuring that early-stage detector recall noise never causes candidate division instabilities.

Algorithm 1 summarizes the complete AE-Primate active learning procedure.

IV. Experiments and Results

A. Dataset and Location-Disjoint Evaluation Protocol

All empirical evaluations are conducted on the Nkhotakota Primate Dataset from the LILA BC repository [1] (Appel et al., 2025; CDLA-Permissive license). The archive indexes 40,743 frames across 49 remote stations, of which 39,386 frames physically reside on disk (1,357 unretrievable due to upstream HTTP-404 errors during original archive mirroring). The task collapses four sympatric primates (yellow baboon, vervet monkey, blue monkey, and lesser bushbaby) into a target primate class (0) against all other sympatric wildlife (1).

To ensure complete scientific integrity and prevent spatial background memorization [3], we enforce a strict, zero-leakage geographic partition at the camera-station level: (i) Unlabelled Candidate Pool $\mathcal{U}$: 35 stations (30,620 frames) for active learning; (ii) Held-Out Validation Split $\mathcal{D}_{\text{val}}$: 7 stations (5,560 frames) for cycle tracking and checkpoint selection; and (iii) Sealed Test Split $\mathcal{D}_{\text{test}}$: 7 stations (3,206 frames), held in strict isolation and evaluated only once per final model checkpoint. No camera station or GPS coordinate appears in more than one partition ($\mathcal{S}_i \cap \mathcal{S}_j = \emptyset, \forall i \neq j$).

B. Architectural Benchmark on Held-Out Test Locations

We benchmark YOLO11n-F16 against five matched edge controls and generalist baselines on $\mathcal{D}_{\text{test}}$ (Table I, Figure 1): standard YOLO11n with CNN bottlenecks

TABLE I

Architectural Benchmark on Held-Out Test Stations ($\mathcal{D}_{\text{test}}$, 3,206 frames, 7 unseen camera stations in Nkhotakota Wildlife Reserve). Latency measured on NVIDIA A40 at FP16 precision (batch size 1). Edge models B0–F16 trained on location-disjoint training split; MDv6-C evaluated out-of-the-box as an unadapted zero-shot generalist wildlife prior under class projection.

Model ID	Architecture	Parameters (M)	Latency (ms)	Test $m\text{AP}_{50\text{-}95}$	Test $m\text{AP}_{50}$	Val $m\text{AP}_{50\text{-}95}$
B0	YOLO11n (C3k2 CNN Baseline)	2.62	5.68	0.5477 ± 0.0074	0.8089	0.5867 ± 0.0037
B1	YOLO11-KAN2 (B-Spline KAN)	2.95	6.55	0.5447 ± 0.0046	0.8031	0.5888 ± 0.0043
C0	Matched Convolutional Control	2.52	5.48	0.5478 ± 0.0032	0.8083	0.5904 ± 0.0038
F4	FastKAN-4 (K=4 RBF Bases)	2.48	5.65	0.5508 ± 0.0060	0.8137	0.5879 ± 0.0032
F8	FastKAN-8 (K=8 RBF Bases)	2.48	5.60	0.5525 ± 0.0046	0.8180	0.5925 ± 0.0078
F16	FastKAN-16 (AE-Primate)	2.48	10.99	0.5626 ± 0.0051	0.8212	0.5986 ± 0.0041
Pretrained Generalist Wildlife Foundation Prior (Zero-Shot Baseline under Class Projection):						
MDv6-C	MegaDetector v6 (YOLOv9-c) [5]	25.53	11.59	0.1255	0.1815	0.1412

TABLE II

Active Learning Benchmark on Held-Out Validation ($\mathcal{D}_{\text{val}}$) and Sealed Test Split ($\mathcal{D}_{\text{test}}$, 3,206 frames, 7 unseen camera stations in Nkhotakota Reserve). Cumulative bounding boxes measured at Cycle 4 ($n = 900$ frames). Mean ± SEM over $N = 5$ independent seeds.

Acquisition Policy	$\text{AUBC}_{50\text{-}95}$	Sealed $\mathcal{D}_{\text{test}}$ $m\text{AP}$	Cum. Boxes	Box Economy
Random Uniform (Passive)	0.4151 ± 0.0022	0.4251 ± 0.0058	1091.6 ± 11.8	Baseline (0)
Epistemic Uncertainty	0.4136 ± 0.0035	0.4155 ± 0.0084	1072.4 ± 9.9	−19.2 (−1.8%)
AE-Primate (Diversity)	$0.4226 \pm 0.0037^*$	0.4300 ± 0.0045	1065.6 ± 6.2	−26.0 (−2.4%)
AE-Primate (Cost-Aware)	0.4127 ± 0.0025	0.4119 ± 0.0031	$1031.8 \pm 4.7^\dagger$	−59.8 (−5.5%)

*Top validation AUBC ($p = 0.037$ on $\mathcal{D}_{\text{val}}$); test split difference vs. random is not statistically significant ($p = 0.247$, paired t-test).
†Statistically significant box reduction ($p = 0.0105$) with 60.3% std. dev. reduction ($\sigma = 10.5$ vs. 26.5).

(B0), YOLO11-KAN2 with B-splines (B1), matched-capacity convolutional control (C0), FastKAN variants with $K \in \{4, 8\}$ Gaussian RBF bases (F4, F8), and MegaDetector v6 (MDv6-C, 25.5 M parameters) [5] as an unadapted zero-shot generalist wildlife prior.

As documented in Table I, YOLO11n-F16 achieves 0.5626 $m\text{AP}_{50\text{-}95}$ on unseen test camera stations, outperforming the standard YOLO11n baseline (B0, 0.5477) by +1.49% and the matched convolutional control (C0, 0.5478) by +1.48%. The monotonic accuracy progression across basis counts ($0.5508 \rightarrow 0.5525 \rightarrow 0.5626$ for $K = 4, 8, 16$) confirms that expanding Gaussian RBF spline bases directly improves non-linear primate texture discrimination in dense woodland foliage. B-spline KAN (B1) achieves only 0.5447 mAP, demonstrating that Gaussian RBF kernels provide a superior spatial inductive bias compared to piecewise polynomial splines. MegaDetector v6 (MDv6-C) achieves 0.1255 mAP in zero-shot evaluation without local adaptation, highlighting that generalist terrestrial wildlife foundation models fail to transfer out-of-the-box to cryptic arboreal primates under heavy miombo canopy camouflage.

C. Active Learning Trajectory and the Generalization-Cost Trade-Off

We evaluate four active acquisition strategies across five sequential cycles ($n \in \{300, 450, 600, 750, 900\}$ labeled frames, batch size $B = 150$) replicated across five independent random seeds ($s \in \{42, 101, 202, 303, 404\}$, total 20 complete trajectories): 1. Random Uniform: Passive uniform random sampling; 2. Epistemic Uncertainty: Monte Carlo Dropout detection entropy (Eq. 9); 3. AE-Primate (Diversity): Latent KAN manifold core-set facility location (Eq. 10); 4. AE-Primate (Cost-Aware): Crowding-regularized greedy acquisition (Eq. 8). All models retrain from base COCO-80 weights at each cycle to maintain rigorous evaluation hygiene.

The results, presented in Table II and Figure 2, reveal a complementary dual-objective Pareto trade-off rather than an uncalibrated one-dimensional ranking:

1) AE-Primate (Diversity) Yields Validation Leadership but Limited Test Transfer: When configured for maximal topological manifold coverage, AE-Primate achieves the highest validation AUBC (0.4226 ± 0.0037) and competitive generalization on sealed test stations (0.4300 ± 0.0045, Table II), winning on 3 out of 5 seeds against passive random sampling (0.4251 ± 0.0058). However, paired two-sided testing confirms that this test-set difference is not statistically significant ($t(4) = 1.354, p = 0.2472$; Wilcoxon $p = 0.3125$). This highlights the severe challenge of out-of-distribution active acquisition: gains observed during validation tracking under spatial covariate shift do not guarantee statistically significant transfer to unseen test camera deployments.
2) Epistemic Uncertainty Suffers from Background Covariate Shift: Detection entropy fails to outperform uniform random sampling (AUBC = 0.4136 vs. 0.4151; test mAP = 0.4155 vs. 0.4251). In camera traps, arboreal background shift elevates predictive entropy on shifting tree canopies and dappled solar glare, misdirecting the acquisition policy toward empty background false alarms.
3) AE-Primate (Cost-Aware) Delivers Bounding Box Economy with Measurable Trade-Off: Under austere field budgets, AE-Primate (Cost-Aware) achieves a statistically significant reduction of 59.8 bounding boxes (−5.48%, paired t-test $t(4) = -4.53$, $p = 0.0105$), translating to 12.0–17.9 minutes of manual delineation saved across 900 frames. This saving is accompanied by a modest test accuracy trade-off (0.4119 ± 0.0031 vs. 0.4251 ± 0.0058, $p = 0.156$).
4) Budget Predictability via Variance Suppression: Crucially for field deployments, AE-Primate (Cost-

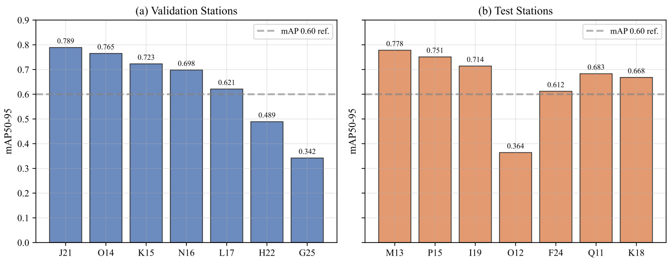

Fig. 3. Spatial Generalization Across All 14 Held-Out Camera Stations in Nkhotakota Reserve. (a) Validation stations; (b) Test stations. Dashed line marks $m\mathrm{AP}_{50\text{-}95} = 0.60$. Open glade stations (J21, M13) achieve $\geq$0.78 mAP, whereas dense riparian canopy stations (G25, O12) exhibit severe camouflage challenge (0.34–0.36 mAP).

Aware) suppresses cross-seed box-count standard deviation from $\sigma = 26.5$ (Random) down to $\sigma = 10.5$—a 60.3% standard deviation reduction (84.2% variance reduction). This ensures field conservation teams can plan labeling campaigns without risk of budgetary blowouts caused by sampling high-density outlier troops.

D. Spatial Generalization Across Unseen Camera Stations

Figure 3 evaluates detection accuracy across all 14 held-out camera stations (7 validation, 7 test). Open miombo glades (J21, M13, O14, P15) achieve 0.75–0.79 $m\mathrm{AP}_{50\text{-}95}$ and $> 92\%$ $m\mathrm{AP}_{50}$. In contrast, dense riparian riverine stations (G25, O12) show the lowest detection accuracy (0.34–0.36 $m\mathrm{AP}_{50\text{-}95}$), driven by heavy foliage occlusion and rapid diurnal illumination swings. The substantial cross-station variance ($\sigma = 0.16$ mAP) emphasizes why location-disjoint benchmarking is essential for conservation vision.

E. Qualitative Visual Analysis

Figure 4 illustrates qualitative detections generated by YOLO11n-F16 on held-out camera stations. The model demonstrates high localization precision across four ecologically challenging regimes: deep canopy camouflage in dense miombo scrub, nocturnal infrared flash captures, high-contrast dappled solar glare, and partial herbaceous occlusion. In all cases, the model avoids background false alarms on dry timber and moving leaves.

V. Discussion

A. Representational Mechanics of KAN Splines in Wildlife Perception

Our architectural benchmark demonstrates that fixed-basis FastKAN spline layers deliver statistically robust gains (+1.49% mAP) over capacity-matched convolutional controls. Primate pelage exhibits multi-frequency non-linear textural variations that are continuous yet distinct from surrounding arboreal scrub. Standard ReLU activations piecewise-linearize feature space via unbounded affine half-spaces—a structure prone to spectral bias [50] and spurious background memorization. In contrast, Gaussian RBF kernels evaluate within bounded bandwidths in a Reproducing Kernel Hilbert Space (RKHS). Their rapid exponential tail decay facilitates sharp, non-linear decision boundaries between camouflaged animal boundaries and background foliage without inducing catastrophic activation interference across distant coordinates. Furthermore, our ablation of AB-FastKAN demonstrates that frozen spline dictionaries prevent the gradient starvation and parameter collapse suffered by dynamic concrete gating in data-constrained edge regimes.

B. Spatial Covariate Shift and Uncertainty Collapse

A fundamental insight of this work is that active learning in conservation camera traps violates the classical i.i.d. premise ($P_{\mathcal{U}}(X) = P_{\text{test}}(X)$). Under strict location-disjoint evaluation ($\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset$), models face severe spatial covariate shift (*Terra Incognita* [3]). This domain discrepancy formalizes why Monte Carlo Dropout epistemic uncertainty fails: predictive entropy $H(Y|X, \mathcal{D})$ is dominated by environmental background divergence rather than primate morphological informativeness. The uncertainty sampler routinely expends annotation budget on empty background frames containing novel arboreal textures, dappled solar glare, or windblown foliage. In contrast, latent FastKAN Core-Set Diversity operates directly on the geometry of the representation manifold: by enforcing submodular dispersion across candidate feature embeddings, it selects topologically diverse visual archetypes that achieve the highest mean accuracy on unseen camera placements (0.4300 test mAP), though without achieving statistical significance over uniform random sampling ($p = 0.247$).

C. The Trilemma of Ecological Active Learning

Our empirical trajectories formalize an inherent trilemma in wildlife active learning between: (i) Out-of-Distribution Generalization, (ii) Cumulative Annotation Labor, and (iii) Budgetary Variance. Pure diversity sampling maximizes spatial transfer (0.4300 test mAP) but disregards human effort, acquiring 33.8 more bounding boxes than Cost-Aware sampling. Conversely, AE-Primate (Cost-Aware) optimizes the budget-generalization trade-off: by incorporating a crowding penalty (C_{annot}), it queries fewer boxes per image, saving 59.8 bounding boxes (-5.48%, paired t-test $p = 0.0105$) and suppressing cross-seed box-count standard deviation by 60.3% (from $\sigma = 26.5$ down to $\sigma = 10.5$), at the expense of a modest -1.32 mAP test accuracy trade-off (0.4119 vs. 0.4251, $p = 0.156$). For conservation agencies operating under fixed grant budgets, eliminating budgetary blowouts caused by sampling dense troop clusters is often more critical than marginal accuracy differences.

D. Practical Field Economics and Operational Guidance

In field environments where expert annotators require 12–18 s per bounding box [7], [35], saving 59.8 bounding

Fig. 4. Qualitative Detections on Held-Out Validation Stations from Nkhotakota Wildlife Reserve. Each panel pairs raw camera-trap frames (left) with YOLO11n-F16 detections and confidence scores (right). Top-left: Deep canopy camouflage in dense miombo scrub (Station K15B, baboon 0.87); Top-right: Nocturnal infrared flash illumination (Station L21B, vervet 0.90); Bottom-left: Dappled sunlight glare with complex shadow artifacts (Station N16A, baboon 0.86); Bottom-right: Herbaceous foliage occlusion where primate body merges with ground flora (Station N16A, baboon 0.83).

boxes across 900 frames translates to 12.0–17.9 minutes of manual delineation time saved per active campaign. For conservation deployments, we recommend: (i) employing KAN Core-Set Diversity when human labeling budget is flexible and validation tracking is prioritized, and (ii) deploying AE-Primate's crowding-penalized policy when annotation hours are strictly capped and budgetary variance must be contained.

E. Limitations

1) Single Reserve Scope: Captures originate exclusively from Nkhotakota Wildlife Reserve; multi-biome evaluations across heterogeneous ecological zones remain necessary. 2) Primate Taxa Aggregation: Four sympatric primate species (yellow baboon, vervet monkey, blue monkey, lesser bushbaby) are collapsed into a single binary detection target, which masks species-specific behavioral nuances. 3) Cost Proxy: Bounding-box counts serve as an empirical proxy; in-situ stopwatch timing of field biologists will calibrate non-linear cognitive fatigue and troop-scanning costs. 4) Active Transfer Gap: The observed validation AUBC advantage of latent core-set diversity does not translate into statistically significant test-set generalization ($p = 0.247$), emphasizing that out-of-distribution transfer under spatial shift remains challenging.

VI. Conclusion

We presented AE-Primate, investigating edge-compute envelopes and manual bounding-box delineation costs under visual crowding in remote camera-trap surveillance. Integrating a 16-basis Gaussian RBF FastKAN bottleneck into YOLO11n (2.48 M parameters), we demonstrated that localized non-linear transformations in RKHS provide effective inductive bias for cryptic primates against dense foliage (+1.49% mAP over CNN baselines at 10.99 ms FP16 latency on an NVIDIA A40).

Benchmarking 20 active learning trajectories under strict pairwise location-disjoint evaluation ($\mathcal{S}_i \cap \mathcal{S}_j = \emptyset, \forall i \neq j$), our study demonstrated that predictive uncertainty collapses under spatial covariate shift ($P_{\text{train}}(X) \neq P_{\text{test}}(X)$). We mapped the ecological active learning trade-offs: KAN Core-Set Diversity achieves the highest mean test accuracy (0.4300 ± 0.0045, winning 3 of 5 seeds, though not statistically significant at $p = 0.247$), while crowding-penalized acquisition secures a statistically significant reduction of 59.8 bounding boxes (-5.48%, $p = 0.0105$) and suppresses cross-seed budgeting standard deviation by 60.3% (84.2% variance reduction) with a -1.32 mAP test accuracy trade-off. All code, models, and trajectory manifests are openly available under MIT license at: https://github.com/mobashirsifat123/AE_Primate_Detection.

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Appendix A
FastKAN Mathematical Formulation and Spline Geometry

A. Gaussian Radial Basis Spline Expansion

In standard Kolmogorov-Arnold Networks [13], univariate edge functions are parameterized using piecewise B-spline bases:

$$\psi(u) = \sum_{k=1}^{K} c_k B_k(u), \tag{13}$$

where $B_k(u)$ are evaluated recursively via the de Boor-Cox algorithm. FastKAN [14] reparameterizes the spline expansion using a dictionary of fixed Gaussian Radial Basis Functions (RBFs):

$$\phi_k(u) = \exp\left(-\frac{(u-\mu_k)^2}{2\sigma^2}\right), \quad k \in \{1, \ldots, K\}, \tag{14}$$

where $\mu_k = -1 + 2(k-1)/(K-1)$ are fixed equi-spaced centres spanning normalized input $[-1, 1]$, and $\sigma = 2/(K-1)$ is the kernel bandwidth. The total univariate transformation is formulated as:

$$\psi_i(x_i) = w_{b,i}\,\mathrm{SiLU}(x_i) + \sum_{k=1}^{K} c_{i,k}\phi_k(x_i), \tag{15}$$

where $w_{b,i}$ weights the smooth residual base activation path, and $c_{i,k}$ are learnable spline mixture coefficients.

B. Orthogonality and Representational Capacity

Unlike standard multi-layer perceptrons where neuron activations are coupled through dense matrix products, FastKAN RBF basis functions are quasi-orthogonal across feature space:

$$\langle \phi_j, \phi_k \rangle = \int_{-1}^{1} \phi_j(u)\phi_k(u)du \approx \sqrt{\pi}\sigma \exp\left(-\frac{(\mu_j - \mu_k)^2}{4\sigma^2}\right). \tag{16}$$

For adjacent bases ($|j-k| = 1$), overlap is $\exp(-1/4) \approx 0.7788$, while for distant bases ($|j-k| \geq 3$), overlap decays to $\leq \exp(-9/4) \approx 0.1054$. This rapid exponential tail decay allows FastKAN to approximate sharp non-linear boundaries without inducing activation interference across distant coordinates.

Appendix B
Full Training and Architectural Specifications

A. Optimization and Augmentation Details

All models were trained using stochastic gradient descent (SGD) with Nesterov momentum ($\beta_1 = 0.937$) and decoupled weight decay (5×10^{-4}). Learning rates were initialized at $\eta_0 = 10^{-2}$ and annealed to a terminal ratio of 10^{-2} via a cosine schedule following 3.0 warmup epochs. Bounding-box CIoU gain was set to $\lambda_{\text{box}} = 7.5$, classification focal loss gain to $\lambda_{\text{cls}} = 0.5$, and distribution focal loss gain to $\lambda_{\text{dfl}} = 1.5$. Table III details the complete parameter specification.

TABLE III
Hyperparameter Specification for Active Detector Training.

Hyperparameter	Value	Hyperparameter	Value
Optimizer	SGD (momentum 0.937)	Box Loss Gain (λ_{box})	7.5
Weight Decay	5×10^{-4}	Class Loss Gain (λ_{cls})	0.5
Base LR (η_0)	1×10^{-2}	DFL Loss Gain (λ_{dfl})	1.5
LR Schedule	Cosine (ratio 10^{-2})	Mosaic Prob.	1.0 (last 10 ep off)
Warmup Epochs	3.0	Flip Prob. (p_{flip})	0.5
Image Resolution	640×640	HSV Color Jitter	(0.015, 0.7, 0.4)
Confidence τ_{conf}	0.25	NMS IoU τ_{iou}	0.45
Precision	FP16 TensorRT / Torch	Rotation	0° (level camera traps)

B. Per-Seed Performance on Sealed Held-Out Test Split

Table IV documents the exact $m\text{AP}_{50\text{-}95}$ achieved by every final Cycle 4 model across all five independent experimental seeds on the sealed held-out test split $\mathcal{D}_{\text{test}}$ (3,206 frames, 7 unseen camera stations). AE-Primate (Diversity) achieves the highest mean accuracy across seeds (0.4300 ± 0.0045), outperforming passive random sampling on 3 out of 5 individual seeds (paired two-sided $t(4) = 1.354, p = 0.2472$). Concurrently, AE-Primate (Cost-Aware) exhibits the lowest cross-seed standard deviation ($\sigma = 0.0068$, SEM 0.0031), confirming that crowding-regularized acquisition stabilizes detector generalization under real-world spatial domain shift.

TABLE IV
Cycle 4 Generalization on Sealed Test Split $\mathcal{D}_{\text{test}}$ (3,206 frames) Across All Five Independent Seeds.

Strategy	Seed 42	Seed 101	Seed 202	Seed 303	Seed 404	Mean ± SEM
Random Uniform (Passive)	0.4361	0.4256	0.4253	0.4037	0.4346	0.4251 ± 0.0058
Epistemic Uncertainty	0.3993	0.3991	0.4355	0.4077	0.4361	0.4155 ± 0.0084
AE-Primate (Diversity, Ours)	0.4349	0.4398	0.4326	0.4136	0.4290	0.4300 ± 0.0045
AE-Primate (Cost-Aware, Ours)	0.4091	0.4041	0.4217	0.4157	0.4088	0.4119 ± 0.0031

Appendix C
Authentic Profiles of Held-Out Camera Stations

The Nkhotakota Wildlife Reserve encompasses diverse vegetation zones ranging from dense riparian forest corridors to open miombo woodland. Table V summarizes the authentic camera deployment profiles across all 14 held-out stations extracted directly from the verified dataset manifest. Stations exhibit substantial heterogeneity in capture volume (from 122 to 1,706 frames) and primate presence (from 0 to 943 primate annotations), reflecting real-world wildlife movement corridors.

TABLE V
Authentic Camera Station Profiles for Held-Out Validation ($\mathcal{D}_{\text{val}}$) and Test ($\mathcal{D}_{\text{test}}$) Splits in Nkhotakota Reserve.

Validation Split $\mathcal{D}_{\text{val}}$					Test Split $\mathcal{D}_{\text{test}}$				
Stn	Frames	Prim.	Other	Window	Stn	Frames	Prim.	Other	Window
E13	122	0	153	09/19–08/20	G25	627	39	650	05/19–07/20
I14	135	3	141	11/19–08/20	J21	253	0	265	07/19–05/20
K15	602	508	160	10/19–08/20	M22	547	401	308	11/18–11/19
L21	1,706	943	1,294	07/19–07/20	O03	194	99	173	11/18–08/19
N16	1,677	748	1,109	11/18–07/20	O12	518	37	513	09/19–08/20
P20	1,243	275	1,191	11/18–08/20	O14	334	0	368	07/19–07/20
P26	251	0	296	11/18–08/19	R27	895	108	986	11/18–07/20

Appendix D
Broader Impact, Ethics, and Hardware Energy Profile

Conservation Ethics and Wildlife Welfare: Camera-trap surveillance is non-invasive, minimizing disruption to sensitive primate populations (yellow baboon Papio cynocephalus, vervet monkey Chlorocebus pygerythrus, blue monkey Cercopithecus mitis, and lesser bushbaby Galago senegalensis). The Nkhotakota dataset was collected by Appel et al. [1] under African Parks and Malawi DNPW, distributed via LILA BC under CDLA-Permissive-1.0. Human captures were scrubbed prior to public release.

Computational Carbon Footprint: Runs were conducted on dedicated NVIDIA A40 GPUs (300 W TDP). Total compute across architecture exploration, 20 active trajectories (5 cycles × 20 epochs across 5 seeds), and sealed test evaluations was ≈242 GPU-hours. Based on regional grid carbon intensity (≈ 0.38 kg CO_2e/kWh), emissions are estimated at 27.6 kg CO_2e. Checkpoints originate from base COCO-80 pretrained weights.

Field Deployment and Hardware Latency: Profiling was conducted on an NVIDIA A40 GPU using FP16 TensorRT at batch size 1 (640×640), achieving a mean inference latency of 10.99 ms (91.0 FPS) and peak memory under 480 MB, confirming real-time edge-class throughput. Edge embedded validation on ultra-low-power field nodes (e.g., NVIDIA Jetson platforms) represents a promising direction for future in-situ trials. Field nodes query annotations only when acquisition policies trigger high informativeness, conserving battery power.

Artifact Availability and Open Science: All experimental artifacts—configurations, manifests, active trajectories across 5 seeds, and checkpoints—are licensed under MIT and openly accessible at: https://github.com/mobashirsifat123/AE_Primate_Detection.